HUNT-SAGE · THREAT HUNTING CHEAT SHEET

Splunk SPL for Hunters

Search patterns for stacking, rarity, time analysis and joins in Splunk.

github.com/iamjammie/hunt-sage · iamjammie.github.io/hunt-sage · Built 2026-10-31

Anatomy

```
index=wineventlog EventCode=4624 Logon_Type=3 earliest=-24h
| search NOT Account_Name="*$"
| stats count dc(ComputerName) as hosts values(Source_Network_Address) as src by Account_Name
| where hosts>5
| sort - hosts
```

Order of operations: **filter in the base search** (index, sourcetype, keywords) → `where` / `search` → transform with `stats`. Field names depend on your Windows add-on and CIM mapping, so always check with `| table *` on a sample first.

Core commands

Command	Use
`stats` / `eventstats` / `streamstats`	Aggregate / aggregate keeping events / running window computations
`tstats`	Very fast counts over indexed fields / data models
`rex field=_raw "(?<name>regex)"`	Extract fields
`regex field=pattern`	Filter by regex
`eval`	Compute: `if`, `case`, `match`, `like`, `len`, `lower`, `split`, `mvindex`
`rare` / `top`	Frequency (rare = long-tail stacking)
`bin _time span=1h`	Bucket time
`timechart`	Time series
`transaction`	Group related events (costly; prefer `stats`)
`lookup` / `inputlookup`	Enrich / allow-lists and watchlists
`join`	Subsearch join (limited; prefer `stats` with `by`)
`append` / `appendcols`	Combine searches
`fillnull`, `dedup`, `sort`, `head`, `table`, `fields`	Housekeeping

Hunting patterns

Stack / long-tail

```
index=sysmon EventCode=1
| stats dc(host) as hosts count by Image
| where hosts<=2
| sort hosts
```

```
index=sysmon EventCode=1 | rare limit=30 Image
```

First-seen / new behaviour

```
index=proxy earliest=-30d
| stats earliest(_time) as first_seen by src_ip dest_domain
| where first_seen > relative_time(now(), "-1d")
```

Beacon timing

```
index=proxy src_ip=* dest_domain=*
| sort 0 src_ip dest_domain _time
| streamstats current=f last(_time) as prev by src_ip dest_domain
| eval gap=_time-prev
| stats count avg(gap) as avg stdev(gap) as sd by src_ip dest_domain
| eval cov=sd/avg
| where count>30 AND avg>10 AND cov<0.35
```

Outliers (z-score) per entity

```
index=firewall action=allowed | bin _time span=1h
| stats sum(bytes_out) as b by src_ip _time
| eventstats avg(b) as mu stdev(b) as sd by src_ip
| eval z=(b-mu)/sd | where z>3
```

Sequence (A then B)

```
index=sysmon (EventCode=1 Image="*\\winword.exe") OR (EventCode=1 ParentImage="*\\winword.exe" (Image="*\\cmd.exe" O
R Image="*\\powershell.exe"))
| transaction host maxspan=5m startswith=(Image="*\\winword.exe") endswith=(ParentImage="*\\winword.exe")
```

Fast counts with tstats

```
| tstats count where index=wineventlog earliest=-24h by host, sourcetype | sort count
| tstats summariesonly=true count from datamodel=Endpoint.Processes where Processes.process_name="rundll32.exe" by P
rocesses.dest Processes.process
```

Regex extraction & decode

```
| rex field=CommandLine "(?i)-enc(odedcommand)?\s+(?<b64>[A-Za-z0-9+/=]{20,})"
| eval decoded=base64decode(b64)    /* use the Base64 app/CyberChef for UTF-16LE */
```

Allow-list via lookup

```
index=sysmon EventCode=1 | lookup approved_parents parent_image AS ParentImage OUTPUT approved
| where isnull(approved)
```

Entropy (URL Toolbox)

```
| `ut_parse_extended(query, ut)` | `ut_shannon(ut_domain)` | where ut_shannon>3.8
```

Performance tips

- Always specify `index=` and a **time range**; use `sourcetype` and indexed fields.
- `tstats` and data-model acceleration for fleet-wide counts.
- Avoid leading wildcards (`*foo`) and `transaction` over long spans.
- Use `| fields` early to drop unneeded fields on big searches.
- Use summary indexing for baselines you reuse.

Common Windows field names (Splunk Add-on for Microsoft Windows)

`EventCode` , `Logon_Type` , `Account_Name` , `Source_Network_Address` , `Authentication_Package` , `Key_Length` , `Workstation_Name` , `Service_Name` , `Ticket_Encryption_Type` , `Task_Name` , `New_Process_Name` , `Process_Command_Line` . Sysmon app: `Image` , `ParentImage` , `CommandLine` , `TargetImage` , `GrantedAccess` , `DestinationIp` .